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(54) **WATER REPELLENT COMPOSITION,
PRODUCTION METHOD FOR
FLUORINE-FREE POLYMER, TREATMENT
METHOD AND ARTICLE**

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(57)

ABSTRACT

The present invention pertains to a water repellent composition containing a non-fluorinated polymer having a structural unit based on a long-chain alkyl (meth)acrylate monomer, a structural unit based on a vinyl halide monomer, and a structural unit based on a monomer having two or more (meth)acryloyl groups.

WATER REPELLENT COMPOSITION, PRODUCTION METHOD FOR FLUORINE-FREE POLYMER, TREATMENT METHOD AND ARTICLE

[0001] This application is a continuation application of International Application No. PCT/JP2023/041929, filed on Nov. 22, 2023, which claims the benefit of priority of the prior Japanese Patent Application No. 2022-186728, filed Nov. 22, 2022 in Japan, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to a water repellent composition, a method for producing a non-fluorinated polymer, a treatment method, and an article.

BACKGROUND ART

[0003] As a method of imparting water repellency to the surface of an article such as a textile product, a method of treating the article using a water repellent composition containing a fluoropolymer has been known. However, the fluoropolymer used in the above method is of concern in terms of high environmental impact. Accordingly, there is a demand for a water repellent composition containing a non-fluorinated polymer that can impart sufficient water repellency to the surface of an article and has a low environmental impact.

[0004] As a water repellent composition containing a non-fluorinated polymer, for example, Patent Document 1 discloses a water repellent composition containing a non-fluorinated copolymer characterized by having a structural unit based on a (meth)acrylate monomer (A) that does not have a polyfluoroalkyl group and has an alkyl group of 20 to 30 carbon atoms, and a structural unit based on a halogenated olefin monomer (B), in which a ratio of the structural unit based on the aforementioned monomer (A) is from 5 to 95% by mass of all structural units (100% by mass), a ratio of the structural unit based on the aforementioned monomer (B) is from 5 to 60% by mass of all structural units (100% by mass), and the total content of the structural unit based on the aforementioned monomer (A) and the structural unit based on the aforementioned monomer (B) is 65% by mass or more of all structural units (100% by mass).

CITATION LIST

Patent Document

[0005] Patent Document 1: International Patent Publication No. 2009/113589

SUMMARY OF INVENTION

Technical Problem

[0006] However, an article treated with the water repellent composition containing a copolymer described in Patent Document 1 may not have sufficient water repellency.

[0007] The present invention has an object of providing: a water repellent composition from which an article with better water repellency can be obtained; a method for producing a non-fluorinated polymer; a treatment method using the aforementioned water repellent composition; and

an article with excellent water repellency that has been treated using the aforementioned water repellent composition.

Solution to Problem

[0008] The present invention includes the following aspects.

[0009] [1] A water repellent composition containing a non-fluorinated polymer having a structural unit based on a long-chain alkyl (meth)acrylate monomer, a structural unit based on a vinyl halide monomer, and a structural unit based on a monomer having two or more (meth)acryloyl groups.

[0010] [2] The water repellent composition according to [1], wherein the aforementioned long-chain alkyl (meth)acrylate monomer is represented by the following formula (1):



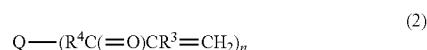
[0011] In the above formula (1), R^1 represents H, CH_3 or a chlorine atom, preferably H or CH_3 , and more preferably H; and R^2 represents an alkyl group having 12 to 30, 14 to 28, or 16 to 24 carbon atoms.

[0012] [3] The water repellent composition according to [2], wherein the aforementioned structural unit based on a long-chain alkyl (meth)acrylate monomer includes a structural unit based on a long-chain alkyl (meth)acrylate monomer in which R^2 in the above formula (1) has 12 to 18, 16 to 18, or 18 carbon atoms, and a structural unit based on a long-chain alkyl (meth)acrylate monomer in which R^2 in the above formula (1) has 19 to 30, 20 to 28, or 20 to 24 carbon atoms.

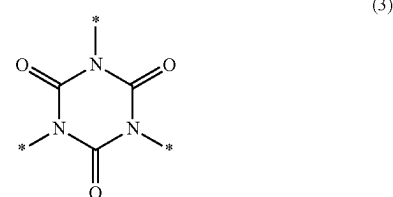
[0013] [4] The water repellent composition according to [3], wherein a ratio of the structural unit based on a long-chain alkyl (meth)acrylate monomer in which R^2 in the above formula (1) has 12 to 18, 16 to 18, or 18 carbon atoms with respect to the total amount of the aforementioned structural unit based on a long-chain alkyl (meth)acrylate monomer is from 1 to 45% by mass, from 5 to 40% by mass, or from 10 to 38% by mass.

[0014] [5] The water repellent composition according to any one of [1] to [4], wherein the aforementioned vinyl halide monomer is vinyl chloride or vinylidene chloride.

[0015] [6] The water repellent composition according to any one of [1] to [5], wherein the aforementioned monomer having two or more (meth)acryloyl groups is represented by the following formula (2):

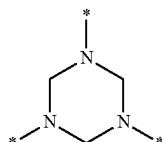


[Chemical Formula 1]



[Chemical Formula 2]

-continued



(4)

[0016] In the above formula (2), n represents an integer of 2 to 10; R^3 represents H, CH_3 or a chlorine atom; R^4 represents a single bond or a divalent organic group; Q represents a linear or branched alkylene group having 1 to 16 carbon atoms, a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, a trivalent group represented by the above formula (3), or a trivalent group represented by the above formula (4); a plurality of $(R^4C(=O)CR^3=CH_2)$ moieties may be the same or different, and * represents a bond in the above formula (3) and the above formula (4), provided that when Q is a linear or branched alkylene group having 1 to 16 carbon atoms or a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, the atom in R^4 bonded to Q is an atom other than an oxygen atom.

[0017] [7] The water repellent composition according to any one of [1] to [6], wherein a ratio of the aforementioned structural unit based on a long-chain alkyl (meth)acrylate monomer is from 50 to 90% by mass, from 55 to 90% by mass, or from 60 to 85% by mass, a ratio of the aforementioned structural unit based on a vinyl halide monomer is from 5 to 30% by mass, from 5 to 25% by mass, or from 8 to 20% by mass, and a ratio of the aforementioned structural unit based on a monomer having two or more (meth)acryloyl groups is from 0.1 to 20% by mass, from 0.5 to 15% by mass, or from 1 to 10% by mass, with respect to the total amount of structural units of the aforementioned non-fluorinated polymer.

[0018] [8] A method for producing a non-fluorinated polymer, the method including polymerizing a mixture containing a long-chain alkyl (meth)acrylate monomer, a vinyl halide monomer, and a monomer having two or more (meth)acryloyl groups in the presence of a surfactant and a polymerization initiator, wherein a ratio of the aforementioned long-chain alkyl (meth)acrylate monomer is from 50 to 90% by mass, from 55 to 90% by mass, or from 60 to 85% by mass, a ratio of the aforementioned vinyl halide monomer is from 5 to 30% by mass, from 5 to 25% by mass, or from 8 to 20% by mass, and a ratio of the aforementioned monomer having two or more (meth)acryloyl groups is from 0.1 to 20% by mass, from 0.5 to 15% by mass, or from 1 to 10% by mass, with respect to the total mass of monomers constituting the aforementioned non-fluorinated polymer.

[0019] [9] The method for producing a non-fluorinated polymer according to [8], further including polymerizing in the presence of a molecular weight modifier.

[0020] [10] The method for producing a non-fluorinated polymer according to [9], wherein a ratio of the aforementioned molecular weight modifier with respect to 100 parts by mass of all monomers constituting the aforementioned non-fluorinated polymer is from 0.1 to 20 parts by mass, from 0.1 to 10 parts by mass, from 0.5 to 9 parts by mass, or from 1 to 8 parts by mass.

[0021] [11] A treatment method using the water repellent composition according to any one of [1] to [7].

[0022] [12] An article treated using the water repellent composition according to any one of [1] to [7].

Advantageous Effects of Invention

[0023] According to the present invention, it is possible to provide: a water repellent composition from which an article with better water repellency can be obtained; a method for producing a non-fluorinated polymer; a treatment method using the aforementioned water repellent composition; and an article with excellent water repellency that has been treated using the aforementioned water repellent composition.

DESCRIPTION OF EMBODIMENTS

[0024] The meanings and definitions of terms in the present specification are as follows.

[0025] A “non-fluorinated polymer” is a generic term for a polymer having a fluorine atom content of 0.1% by mass or less with respect to the total mass of the polymer. The fluorine atom content with respect to the total mass of the polymer can be measured by combustion ion chromatography or the like.

[0026] A “structural unit based on a monomer” is a generic term for an atomic group directly formed by polymerization of a single monomer molecule and an atomic group obtained by chemical conversion of a portion of this directly formed atomic group.

[0027] A “(meth)acrylate” is a generic term for an acrylate, a methacrylate, and a compound obtained by substituting a methyl group in a methacryloyl group of the methacrylate with a chlorine atom. Similarly, a “(meth)acryloyl group” is a generic term for an acryloyl group, a methacryloyl group, and a group obtained by substituting a methyl group in a methacryloyl group with a chlorine atom.

[0028] A “long-chain alkyl” refers to an alkyl group having 12 or more carbon atoms.

[0029] A “vinyl halide” is a compound in which at least one hydrogen atom in ethylene is substituted with a halogen atom.

[0030] The number average molecular weight (hereinafter also referred to as “ M_n ”) and weight average molecular weight (hereinafter also referred to as “ M_w ”) of a polymer are polystyrene equivalent molecular weights obtained through measurement by gel permeation chromatography (hereinafter also referred to as “GPC”) using a calibration curve produced using a standard polystyrene sample.

[0031] A solid content concentration is calculated by a formula: $(\text{solid content mass})/(\text{sample mass}) \times 100$ by taking the mass of a sample before heating as the sample mass, and the mass of the sample after drying for 4 hours in a convective dryer at 120° C. as the solid content mass.

[0032] A symbol “-” indicating a numerical range means that numerical values described before and after this symbol are included as the lower limit value and the upper limit value.

<<Water Repellent Composition>>

[0033] A water repellent composition of the present embodiment (hereinafter also referred to as “the present composition”) contains a specific polymer (hereinafter also referred to as a “polymer (A)”). The polymer (A) is a

non-fluorinated polymer having a structural unit based on a long-chain alkyl (meth)acrylate monomer (hereinafter also referred to as a “monomer (a)”), a structural unit based on a vinyl halide monomer (hereinafter also referred to as a “monomer (b)”), and a structural unit based on a monomer having two or more (meth)acryloyl groups (hereinafter also referred to as a “monomer (c)”). The polymer (A) may contain either one or both of a structural unit based on a monomer (d) and a structural unit based on a monomer (e) described below.

[0034] The present composition may contain the polymer (A) and a medium.

[0035] The present composition may contain one or more selected from the group consisting of a surfactant, a molecular weight modifier, and a polymerization initiator.

[0036] The present composition may contain other components, if necessary.

[0037] The present composition is preferably a non-fluorinated polymer solution containing the polymer (A). The non-fluorinated polymer solution also includes a dispersion liquid obtained by the production method described below, and a dispersion liquid obtained by further dilution with an arbitrary medium for treating an article.

[0038] The present composition may be a non-fluorinated polymer solution containing the polymer (A) and an organic solvent as a medium, and containing no surfactant.

<Polymer (A)>

[0039] The polymer (A) has a unit based on the monomer (a) (hereinafter also referred to as a “unit (a)”), a unit based on the monomer (b) (hereinafter also referred to as a “unit (b)”), and a unit based on the monomer (c) (hereinafter also referred to as a “unit (c)”).

[0040] The polymer (A) may further have either one or both of a unit based on the monomer (d) (hereinafter also referred to as a “unit (d)”) and a unit based on the monomer (e) (hereinafter also referred to as a “unit (e)”), as necessary.

(Monomer (a))

[0041] The monomer (a) is a long-chain alkyl (meth)acrylate monomer. The long-chain alkyl (meth)acrylate is a monomer having an alkyl group with 12 or more carbon atoms and one (meth)acryloyl group in one molecule.

[0042] As the monomer (a), a long-chain alkyl (meth)acrylate monomer represented by the following formula (1) is preferred.



[0043] In the above formula (1), R^1 represents H, CH_3 or a chlorine atom, and R^2 represents an alkyl group having 12 to 30 carbon atoms.

[0044] R^1 is preferably H or CH_3 , and more preferably H.

[0045] The long-chain alkyl group may be linear or branched, and is preferably linear.

[0046] R^2 preferably has 12 to 30 carbon atoms, more preferably 14 to 28 carbon atoms, and still more preferably 16 to 24 carbon atoms. When the number of carbon atoms is equal to or more than the above lower limit value, the water repellency is excellent. When the number of carbon atoms is equal to or less than the above upper limit value, the availability and handling properties are excellent.

[0047] As the monomer (a), lauryl (meth)acrylate, cetyl (meth)acrylate, stearyl (meth)acrylate, and behenyl (meth)acrylate are preferred.

[0048] Further, two or more types of monomers (a) may be used in combination. As the monomer (a), it is particularly preferable to use a long-chain alkyl (meth)acrylate monomer in which R^2 in the above formula (1) has 12 to 18 carbon atoms (hereinafter also referred to as a “monomer (a1)”) and a long-chain alkyl (meth)acrylate monomer in which R^2 in the above formula (1) has 19 to 30 carbon atoms (hereinafter also referred to as a “monomer (a2)”) in combination.

[0049] The monomer (a1) may be a mixture of any two or more of the long-chain alkyl (meth)acrylate monomers in which R^2 in the above formula (1) has 12 to 18 carbon atoms. The monomer (a2) may be a mixture of any two or more of the long-chain alkyl (meth)acrylate monomers in which R^2 in the above formula (1) has 19 to 30 carbon atoms.

[0050] The number of carbon atoms in the alkyl group of the monomer (a1) is from 12 to 18, preferably from 16 to 18, and more preferably 18. When the number of carbon atoms is equal to or more than the above lower limit value, the water repellency is excellent. When the number of carbon atoms is equal to or less than the above upper limit value, the availability and handling properties are excellent.

[0051] The number of carbon atoms in the alkyl group of the monomer (a2) is from 19 to 30, preferably from 20 to 28, and more preferably from 20 to 24. When the number of carbon atoms is equal to or more than the above lower limit value, the water repellency is excellent. When the number of carbon atoms is equal to or less than the above upper limit value, the availability and handling properties are excellent.

[0052] When the monomer (a1) and the monomer (a2) are used in combination, the ratio of the structural unit based on the monomer (a1) with respect to the total amount of the structural units based on the monomer (a) is preferably from 1 to 45% by mass, more preferably from 5 to 40% by mass, and still more preferably from 10 to 38% by mass. When the above ratio is equal to or more than the above lower limit value, the initial water repellency is further improved. When the above ratio is equal to or less than the above upper limit value, the water repellency after washing is further improved.

[0053] When the monomer (a1) and the monomer (a2) are used in combination, the ratio of the structural unit based on the monomer (a2) with respect to the total amount of the structural units based on the monomer (a) is preferably from 55 to 99% by mass, more preferably from 60 to 95% by mass, and still more preferably from 62 to 90% by mass. When the above ratio is equal to or more than the above lower limit value, the initial water repellency is further improved. When the above ratio is equal to or less than the above upper limit value, the water repellency after washing is further improved.

[0054] When the monomer (a1) and the monomer (a2) are used in combination, the ratio of the structural unit based on the monomer (a1) and the structural unit based on the monomer (a2) with respect to the total amount of the structural units based on the monomer (a) is preferably 70% by mass or more, more preferably 72% by mass or more, still more preferably 75% by mass or more, particularly preferably 85% by mass or more, and may be 100% by mass. When the above ratio is equal to or more than the above lower limit value, the initial water repellency is further improved. When the above ratio is equal to or less than the above upper limit value, the water repellency after washing is further improved.

(Monomer (b))

[0055] The monomer (b) is a vinyl halide. As the monomer (b), a vinyl halide having one or two halogen atoms is preferred, and a vinylidene halide having two halogen atoms is more preferred.

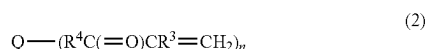
[0056] The halogen atom contained in the above vinyl halide is preferably a chlorine atom, a bromine atom, or an iodine atom, more preferably a chlorine atom or a bromine atom, and still more preferably a chlorine atom. Further, when the above vinyl halide contains two or more halogen atoms, a plurality of halogen atoms may be the same or different.

[0057] The monomer (b) is preferably vinyl chloride or vinylidene chloride.

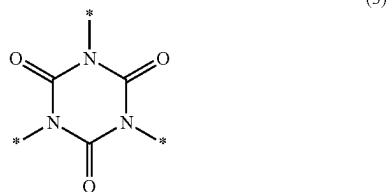
[0058] Two or more types of monomers (b) may be used in combination.

(Monomer (c))

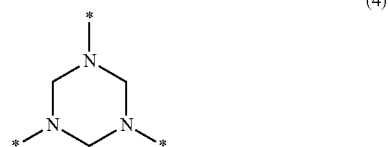
[0059] The monomer (c) is a monomer having two or more (meth)acryloyl groups. As the monomer (c), a monomer having two or more (meth)acryloyl groups represented by the following formula (2) is preferred.



[Chemical Formula 3]



[Chemical Formula 4]



[0060] In the above formula (2), n represents an integer of 2 to 10; R³ represents H, CH₃ or a chlorine atom; R⁴ represents a single bond or a divalent organic group; Q represents a linear or branched alkylene group having 1 to 16 carbon atoms, a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, a trivalent group represented by the above formula (3), or a trivalent group represented by the above formula (4); a plurality of (R⁴C(=O)CR³=CH₂) moieties may be the same or different, and * represents a bond in the above formula (3) and the above formula (4). However, when Q is a linear or branched alkylene group having 1 to 16 carbon atoms or a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, the atom in R⁴ bonded to Q is an atom other than an oxygen atom.

[0061] n is an integer of 2 to 10, preferably an integer of 2 to 8, and more preferably an integer of 2 to 6.

[0062] R³ represents H, CH₃, or a chlorine atom. R³ is preferably H or CH₃.

[0063] R⁴ represents a single bond or a divalent organic group. When R⁴ is a divalent organic group, the number of carbon atoms in R⁴ is preferably from 2 to 12, more preferably from 2 to 8, and still more preferably from 1 to 6.

[0064] When Q is a linear or branched alkylene group having 1 to 16 carbon atoms or a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, it is preferable that R⁴ does not have an oxygen atom.

[0065] When R⁴ is a single bond or an alkylene group, Q is a linear or branched alkylene group having 1 to 16 carbon atoms, and the total number of carbon atoms in Q and R⁴ is 16 or less, Q is a linear or branched alkylene group having 1 to 16 carbon atoms and R⁴ is a single bond.

[0066] When Q is a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol or a trivalent group represented by the above formula (4), R⁴ is preferably a single bond. When Q is a linear or branched alkylene group having 1 to 16 carbon atoms or a trivalent group represented by the above formula (3), R⁴ is preferably a divalent organic group.

[0067] When R⁴ is a divalent organic group, R⁴ is preferably a divalent hydrocarbon group, a divalent organic group in which one or more carbon atoms in a divalent hydrocarbon group are substituted with a hetero atom, or an organic group containing no carbon atoms.

[0068] Examples of the divalent hydrocarbon group include a divalent saturated hydrocarbon group and a divalent unsaturated hydrocarbon group. Examples of the divalent unsaturated hydrocarbon group include a group in which one or more carbon-carbon single bonds in a divalent saturated hydrocarbon group have been replaced with carbon-carbon double bonds or triple bonds.

[0069] Examples of the hetero atom include a nitrogen atom, an oxygen atom, and a sulfur atom, and a nitrogen atom or an oxygen atom is preferred. When R⁴ contains a hetero atom, the number of hetero atoms is preferably from 1 to 4, and more preferably from 1 to 3.

[0070] R⁴ is linear or branched, and is preferably linear.

[0071] When R⁴ is a divalent organic group, R⁴ is preferably a divalent group represented by the following formula (5) or (6).



[0072] In the above formulas (5) and (6), each of R⁵ to R⁸ is independently a single bond or an alkylene group having 1 to 6 carbon atoms. R⁵ is bonded to Q, and R⁶ is bonded to C(=O). R⁷ is bonded to Q, and R⁸ is bonded to C(=O). The alkylene group may be linear or branched, and is preferably linear.

[0073] The alkylene group preferably has 1 to 4 carbon atoms, more preferably 1 to 3 carbon atoms, and still more preferably 1 or 2 carbon atoms.

[0074] In the above formula (5), a divalent group in which R⁵ and R⁶ are single bonds is preferred.

[0075] In the above formula (6), a divalent group in which R⁷ is an alkylene group having 1 to 6 carbon atoms, and R⁸ is a single bond is preferred.

[0076] When Q is a linear or branched alkylene group having 1 to 16 carbon atoms, the number of carbon atoms in

Q is preferably from 1 to 8, more preferably from 1 to 6, and still more preferably from 1 to 3. The alkylene group is preferably linear.

[0077] Examples of the alkylene group include a methylene group, an ethylene group, a propylene group, an isopropylene group, and a butylene group.

[0078] When Q is a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, the number of hydroxyl groups in the polyhydric alcohol is preferably from 2 to 10, more preferably from 2 to 8, and still more preferably from 2 to 6. The polyhydric alcohol may be linear or branched.

[0079] The number of carbon atoms in the polyhydric alcohol is preferably from 2 to 20, more preferably from 2 to 18, and still more preferably from 3 to 15.

[0080] Examples of the polyhydric alcohol include a compound in which 2 to 10 hydrogen atoms in a hydrocarbon compound that may have an etheral oxygen atom are substituted with hydroxyl groups. The above hydrocarbon compound may be a saturated hydrocarbon compound or an unsaturated hydrocarbon compound. Examples of the unsaturated hydrocarbon compound include a compound in which one or more carbon-carbon single bonds in the above saturated hydrocarbon compound are substituted with carbon-carbon double bonds or triple bonds, and a compound having an aromatic ring. The aromatic ring may be included as an aryl group or an arylene group. When the polyhydric alcohol is a compound having an aromatic ring, the number of aromatic rings is preferably 1 or 2.

[0081] When the polyhydric alcohol has an etheral oxygen atom, the number of etheral oxygen atoms is preferably from 1 to 5, more preferably from 1 to 4, and still more preferably from 1 to 3.

[0082] Examples of the monomer (c) include 1,4-butanediol diacrylate, 1,4-butanediol dimethacrylate, 1,6-hexanediol diacrylate, 1,6-hexanediol dimethacrylate, 1,10-decanediol diacrylate, 1,10-decanediol dimethacrylate, trimethylolpropane triacrylate, trimethylolpropane trimethacrylate, ditrimethylolpropane tetraacrylate, ditrimethylolpropane tetramethacrylate, dipentaerythritol hexaacrylate, dipentaerythritol hexamethacrylate, bisphenol A diacrylate, bisphenol A dimethacrylate, tris(2-acryloyloxyethyl) isocyanurate, tris(2-methacryloyloxyethyl) isocyanurate, 1,3,5-triacryloylhexahydro-1,3,5-triazine, 1,3,5-trimethacryloylhexahydro-1,3,5-triazine, N,N'-methylenebisacrylamide, and N,N'-methylenebismethacrylamide.

[0083] Two or more types of monomers (c) may be used in combination.

(Monomer (d))

[0084] The monomer (d) is a monomer having a crosslinkable functional group and one group polymerizable with a (meth)acryloyl group, or a monomer having two or more groups polymerizable with a (meth)acryloyl group (excluding a monomer having two or more (meth)acryloyl groups).

[0085] When the polymer (A) has a structural unit based on the monomer (d), the washing and friction durability are further improved.

[0086] Examples of the group polymerizable with a (meth)acryloyl group include a group having a carbon-carbon double bond at the molecular end, and for example, a (meth)acryloyl group, a vinyl group, or an allyl group is preferred.

[0087] As the crosslinkable functional group, a functional group having at least one bond of a covalent bond, an ionic bond, or a hydrogen bond, or a functional group capable of forming a crosslinked structure through the interaction of the above bonds is preferred.

[0088] As the above functional group, an isocyanate group, a blocked isocyanate group, an alkoxyisilyl group, a primary amino group, an alkoxyethylamide group, a silanol group, a primary amide group, an epoxy group, a hydroxyl group, an oxazoline group, a carboxyl group, a sulfonic acid group, or the like is preferred, and a hydroxyl group, a blocked isocyanate group, a primary amino group, or an epoxy group is particularly preferred.

[0089] As the monomer (d), (meth)acrylates having a crosslinkable functional group, (meth)acrylamides, vinyl ethers having a crosslinkable functional group, or vinyl esters having a crosslinkable functional group are preferred.

[0090] Examples of the monomer (d) include the following compounds.

[0091] 2-isocyanatoethyl (meth)acrylate, 3-isocyanatopropyl (meth)acrylate, 4-isocyanatobutyl (meth)acrylate, a 2-butanone oxime adduct of 2-isocyanatoethyl (meth)acrylate, a pyrazole adduct of 2-isocyanatoethyl (meth)acrylate, a 3,5-dimethylpyrazole adduct of 2-isocyanatoethyl (meth)acrylate, a 3-methylpyrazole adduct of 2-isocyanatoethyl (meth)acrylate, an F-caprolactam adduct of 2-isocyanatoethyl (meth)acrylate, a 2-butanone oxime adduct of 3-isocyanatopropyl (meth)acrylate, and a pyrazole adduct of 3-isocyanatopropyl (meth)acrylate.

[0092] A 3,5-dimethylpyrazole adduct of 3-isocyanatopropyl (meth)acrylate, a 3-methylpyrazole adduct of 3-isocyanatopropyl (meth)acrylate, an F-caprolactam adduct of 3-isocyanatopropyl (meth)acrylate, a 2-butanone oxime adduct of 4-isocyanatobutyl (meth)acrylate, a pyrazole adduct of 4-isocyanatobutyl (meth)acrylate, a 3,5-dimethylpyrazole adduct of 4-isocyanatobutyl (meth)acrylate, a 3-methylpyrazole adduct of 4-isocyanatobutyl (meth)acrylate, and an F-caprolactam adduct of 4-isocyanatobutyl (meth)acrylate.

[0093] Methoxymethyl (meth)acrylamide, ethoxymethyl (meth)acrylamide, butoxymethyl (meth)acrylamide, diacetone acrylamide, γ -methacryloyloxypropyltrimethoxysilane, trimethoxyvinylsilane, and vinyltrimethoxysilane.

[0094] t-butyl (meth)acrylamide sulfonic acid, (meth)acrylamide, N-methylol (meth)acrylamide, N-butoxymethyl (meth)acrylamide, diacetone (meth)acrylamide, glycidyl (meth)acrylate, 2-hydroxyethyl (meth)acrylate, 2-hydroxypropyl (meth)acrylate, 4-hydroxybutyl (meth)acrylate, 3-chloro-2-hydroxypropyl methacrylate, polyoxyalkylene glycol mono(meth)acrylate, (meth)acrylic acid, 2-(meth)acryloyloxyethyl succinic acid, 2-(meth)acryloyloxyhexahydrophthalic acid, 2-(meth)acryloyloxyethyl acid phosphate, allyl (meth)acrylate, 2-vinyl-2-oxazoline, and a polycaprolactone ester of 2-vinyl-4-methyl-(2-vinyl-oxazoline) hydroxyethyl (meth)acrylate.

[0095] Tri(meth)allyl isocyanurate (T(M)AIC, manufactured by Nippon Kasei Chemical Co., Ltd.), triallyl cyanurate (TAC, manufactured by Nippon Kasei Chemical Co., Ltd.), and 3-(methyl ethyl ketoxime)isocyanatomethyl-3,5,5-trimethylcyclohexyl(2-hydroxyethyl methacrylate) cyanate (Techcoat HE-6P, manufactured by Kyoken Kasei). A polycaprolactone ester of hydroxyethyl(meth)acrylate (Placel FA, FM series, manufactured by Daicel Chemical Industries, Ltd.).

[0096] As the monomer (d), N-methylol (meth)acrylamide, 2-hydroxyethyl (meth)acrylate, glycidyl (meth)acrylate, 3-chloro-2-hydroxypropyl methacrylate or a polycaprolactone ester of hydroxyethyl (meth)acrylate (Placel FA, FM series, manufactured by Daicel Chemical Industries, Ltd.), and a 3,5-dimethylpyrazole adduct of 2-isocyanatoethyl (meth)acrylate are preferred.

[0097] Two or more types of monomers (d) may be used in combination.

(Monomer (e))

[0098] The monomer (e) is a monomer excluding the monomer (a), the monomer (b), the monomer (c), and the monomer (d).

[0099] Examples of the monomer (e) include the following compounds.

[0100] Methyl (meth)acrylate, ethyl (meth)acrylate, propyl (meth)acrylate, butyl (meth)acrylate, cyclohexyl (meth)acrylate, 2-ethylhexyl (meth)acrylate, n-hexyl (meth)acrylate, benzyl (meth)acrylate, octyl (meth)acrylate, decyl methacrylate, cyclododecyl acrylate, 3-ethoxypropyl acrylate, methoxy-butyl acrylate, 2-ethylbutyl acrylate, 1,3-dimethylbutyl acrylate, and 2-methylpentyl acrylate.

[0101] Vinyl acetate, vinyl propionate, butene, isoprene, butadiene, ethylene, propylene, vinyl ethylene, pentene, ethyl-2-propylene, butylethylene, cyclohexylpropylethylene, decylethylene, dodecylethylene, hexene, isohexylethylene, neopentylethylene, (1,2-diethoxycarbonyl)ethylene, (1,2-dipropoxycarbonyl)ethylene, methoxyethylene, ethoxyethylene, butoxyethylene, 2-methoxypropylene, pentyloxyethylene, cyclopentanoxyethylene, cyclopentylacetoxyethylene, styrene, α -methylstyrene, p-methylstyrene, hexylstyrene, octylstyrene, and nonylstyrene.

[0102] N-methyl (meth)acrylamide, N,N-dimethyl (meth)acrylamide, N,N-diethyl (meth)acrylamide, N,N-dimethylaminoethyl (meth)acrylamide, N,N-dimethylaminopropyl (meth)acrylamide, N,N-diethylaminoethyl (meth)acrylamide, N,N-diethylaminopropyl (meth)acrylamide, dimethylaminoethyl (meth)acrylate, diethylaminoethyl (meth)acrylate, dimethylaminopropyl (meth)acrylate, (meth)acryloyloxyethyl trimethylammonium chloride, (meth)acryloyloxypropyl trimethylammonium chloride, (meth)acrylamideethyl trimethylammonium chloride, (meth)acrylamidepropyl trimethylammonium chloride, and (meth)acryloyl morpholine.

[0103] A vinyl alkyl ether, a halogenated alkyl vinyl ether, a vinyl alkyl ketone, aziridinyl ethyl (meth)acrylate, a 2-ethylhexyl polyoxyalkylene (meth)acrylate, and a polyoxyalkylene di(meth)acrylate.

[0104] A crotonic acid alkyl ester, a maleic acid alkyl ester, a fumaric acid alkyl ester, a citraconic acid alkyl ester, a mesaconic acid alkyl ester, triallyl cyanurate, allyl acetate, N-vinylcarbazole, maleimide, N-methyl maleimide, a (meth)acrylate having silicone within a side chain, a (meth)acrylate having a urethane bond, a (meth)acrylate having a polyoxyalkylene chain whose terminal is an alkyl group having 1 to 4 carbon atoms, and an alkylene di(meth)acrylate.

[0105] Two or more types of monomers (e) may be used in combination.

[0106] A total ratio of the unit (a), the unit (b) and the unit (c) with respect to all units constituting the polymer (A) is preferably 70% by mass or more, more preferably from 75 to 98% by mass, and still more preferably from 80 to 97%

by mass. When the total ratio of the unit (a), the unit (b) and the unit (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved.

[0107] When the polymer (A) contains either one or both of the unit (d) and the unit (e), the total ratio of the unit (a), the unit (b) and the unit (c) with respect to all units constituting the polymer (A) is preferably from 75 to 98% by mass, and more preferably from 80 to 97% by mass.

[0108] A ratio of the unit (a) with respect to all units constituting the polymer (A) is preferably from 50 to 90% by mass, more preferably from 55 to 90% by mass, and still more preferably from 60 to 85% by mass. When the ratio of the unit (a) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the unit (a) is equal to or less than the above upper limit value, the unit (b), the unit (c) and other units can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0109] A ratio of the unit (b) with respect to all units constituting the polymer (A) is preferably from 5 to 30% by mass, more preferably from 5 to 25% by mass, and still more preferably from 8 to 20% by mass. When the ratio of the unit (b) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the unit (b) is equal to or less than the above upper limit value, the unit (a), the unit (c) and other units can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0110] A ratio of the unit (c) with respect to all units constituting the polymer (A) is preferably from 0.1 to 20% by mass, more preferably from 0.5 to 15% by mass, and still more preferably from 1 to 10% by mass. When the ratio of the unit (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the unit (c) is equal to or less than the above upper limit value, the unit (a), the unit (b) and other units can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0111] A total ratio of the unit (a) and the unit (b) with respect to all units constituting the polymer (A) is preferably 55% by mass or more, more preferably from 60 to 98% by mass, and still more preferably from 68 to 96% by mass. When the total ratio of the unit (a) and the unit (b) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the total ratio of the unit (a) and the unit (b) is equal to or less than the above upper limit value, the unit (c) and other units can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0112] A total ratio of the unit (b) and the unit (c) with respect to all units constituting the polymer (A) is preferably from 5.1 to 50% by mass, more preferably from 5.5 to 40% by mass, and still more preferably from 6 to 30% by mass. When the total ratio of the unit (b) and the unit (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the total ratio of the unit (b) and the unit (c) is equal to or less than the above upper limit value, the unit (a) and other units can be sufficiently con-

tained, and the water repellency of the article treated with the present composition is further improved.

[0113] A total ratio of the unit (c) and the unit (a) with respect to all units constituting the polymer (A) is preferably from 50.1 to 95% by mass, more preferably from 55 to 90% by mass, and still more preferably from 60 to 90% by mass. When the total ratio of the unit (c) and the unit (a) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the total ratio of the unit (c) and the unit (a) is equal to or less than the above upper limit value, the unit (b) and other units can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0114] A mass ratio of unit (a)/unit (b) is preferably from 1.6 to 19, more preferably from 2.2 to 18, and still more preferably from 3 to 10. When the mass ratio of unit (a)/unit (b) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of unit (a)/unit (b) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0115] A mass ratio of unit (b)/unit (c) is preferably from 0.25 to 300, more preferably from 0.3 to 50, and still more preferably from 0.8 to 20. When the mass ratio of unit (b)/unit (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of unit (b)/unit (c) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0116] A mass ratio of unit (c)/unit (a) is preferably from 0.001 to 0.4, more preferably from 0.005 to 0.27, and still more preferably from 0.010 to 0.17. When the mass ratio of unit (c)/unit (a) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of unit (c)/unit (a) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0117] A mass ratio of (unit (a)+unit (b))/unit (c) is preferably from 2.75 to 950, more preferably from 4 to 800, and still more preferably from 6 to 200. When the mass ratio of (unit (a)+unit (b))/unit (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of (unit (a)+unit (b))/unit (c) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0118] When the polymer (A) contains the unit (d), a ratio of the unit (d) with respect to all units constituting the polymer (A) is preferably from 0.1 to 15% by mass, more preferably from 0.2 to 10% by mass, and still more preferably from 0.5 to 6% by mass. When the ratio of the unit (d) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the unit (d) is equal to or less than the above upper limit value, the unit (a), the unit (b), the unit (c) and other units can be

sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0119] When the polymer (A) contains the unit (e), a ratio of the unit (e) with respect to all units constituting the polymer (A) is preferably from 0.1 to 15% by mass, more preferably from 0.2 to 10% by mass, and still more preferably from 0.5 to 6% by mass. When the ratio of the unit (e) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the unit (e) is equal to or less than the above upper limit value, the unit (a), the unit (b), the unit (c) and other units can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0120] When the polymer (A) contains the unit (d) or the unit (e), a total ratio of the unit (d) and the unit (e) with respect to all units constituting the polymer (A) is preferably from 0.2 to 30% by mass, more preferably from 0.4 to 20% by mass, and still more preferably from 1 to 12% by mass. When the ratio of the unit (d) and the unit (e) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the unit (d) and the unit (e) is equal to or less than the above upper limit value, the unit (a), the unit (b) and the unit (c) can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0121] The ratio of each unit can be calculated from the reaction rate of each monomer component by ¹H-NMR, gas chromatography, and high performance liquid chromatography. In a case where the conversion rate of the monomer component to the polymer (A) is high (for example, 90% or more) during the production of the polymer (A), the ratio of each unit may be calculated based on the charged amount of the monomer component.

[0122] The Mn of the polymer (A) is preferably from 5,000 to 100,000, more preferably from 8,000 to 90,000, and still more preferably from 10,000 to 60,000. When the Mn of the polymer (A) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the Mn of the polymer (A) is equal to or less than the above upper limit value, the water dispersibility of the polymer (A) is further improved.

[0123] The Mw of the polymer (A) is preferably from 8,000 to 1,000,000, more preferably from 12,000 to 900,000, and still more preferably from 15,000 to 800,000. When the Mw of the polymer (A) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the Mw of the polymer (A) is equal to or less than the above upper limit value, the water dispersibility of the polymer (A) is further improved.

(Medium)

[0124] Examples of the medium include water, an alcohol, a glycol, a glycol ether, a halogen compound, a hydrocarbon, a ketone, an ester, an ether, a nitrogen compound, a sulfur compound, an inorganic solvent, and an organic acid. Among these, from the viewpoints of solubility and ease of handling, one or more media selected from the group consisting of water, an alcohol, a glycol, a glycol ether, and a glycol ester are preferred.

[0125] Examples of the alcohol include methanol, ethanol, 1-propanol, 2-propanol, 1-butanol, 2-butanol, 2-methylpropanol, 1,1-dimethylethanol, 1-pentanol, 2-pentanol, 3-pentanol, 2-methyl-1-butanol, 3-methyl-1-butanol, 1,1-dimethylpropanol, 3-methyl-2-butanol, 1,2-dimethylpropanol, 1-hexanol, 2-methyl-1-pentanol, 4-methyl-2-pentanol, 2-ethyl-1-butanol, 1-heptanol, 2-heptanol, and 3-heptanol.

[0126] Examples of the glycol or glycol ether include ethylene glycol, ethylene glycol monomethyl ether, ethylene glycol monoethyl ether, ethylene glycol monobutyl ether, ethylene glycol monomethyl ether acetate, ethylene glycol monoethyl ether acetate, propylene glycol, propylene glycol monomethyl ether, propylene glycol monoethyl ether, propylene glycol dimethyl ether, dipropylene glycol, dipropylene glycol monomethyl ether, dipropylene glycol dimethyl ether, dipropylene glycol monoethyl ether, tripropylene glycol, tripropylene glycol monomethyl ether, polypropylene glycol, and hexylene glycol.

[0127] Examples of the halogen compound include a halogenated hydrocarbon and a halogenated ether. It is preferable that the halogen compound does not contain fluorine. Examples of the halogenated hydrocarbon include a hydrochlorocarbon, and a hydrobromocarbon.

[0128] Examples of the hydrocarbon include an aliphatic hydrocarbon, an alicyclic hydrocarbon, and an aromatic hydrocarbon.

[0129] Examples of the aliphatic hydrocarbon include pentane, 2-methylbutane, 3-methylpentane, hexane, 2,2-dimethylbutane, 2,3-dimethylbutane, heptane, octane, 2,2,4-trimethylpentane, 2,2,3-trimethylhexane, decane, undecane, dodecane, 2,2,4,6,6-pentamethylheptane, tridecane, tetradecane, and hexadecane.

[0130] Examples of the alicyclic hydrocarbon include cyclopentane, methylcyclopentane, cyclohexane, methylcyclohexane, and ethylcyclohexane.

[0131] Examples of the aromatic hydrocarbon include benzene, toluene, and xylene.

[0132] Examples of the ketone include acetone, methyl ethyl ketone, 2-pentanone, 3-pentanone, 2-hexanone, and methyl isobutyl ketone.

[0133] Examples of the ester include methyl acetate, ethyl acetate, butyl acetate, methyl propionate, methyl lactate, ethyl lactate, and pentyl lactate.

[0134] Examples of the ether include diisopropyl ether, dioxane, and tetrahydrofuran.

[0135] Examples of the nitrogen compound include pyridine, N,N-dimethylformamide, N,N-dimethylacetamide, and N-methylpyrrolidone.

[0136] Examples of the sulfur compound include dimethyl sulfoxide and sulfolane.

[0137] Examples of the inorganic solvent include liquid carbon dioxide.

[0138] Examples of the organic acid include acetic acid, propionic acid, malic acid, and lactic acid.

[0139] A single type of medium may be used alone, or two or more types thereof may be mixed and used in combination. When two or more types of media are mixed and used in combination, it is preferable to use them by mixing with water. By using a mixed medium, it is easy to control the solubility and dispersibility of the polymer, and it is easy to control the permeability, wettability, a solvent drying speed, and the like with respect to an article during processing.

[0140] When the medium is a mixture of an organic solvent and water, the content of the organic solvent is preferably from 1 to 40 parts by mass, more preferably from 2 to 30 parts by mass, and still more preferably from 3 to 25 parts by mass with respect to 100 parts by mass of water.

(Surfactant)

[0141] Examples of the surfactant include a hydrocarbon-based surfactant. Examples of the hydrocarbon-based surfactant include an anionic surfactant, a nonionic surfactant, a cationic surfactant, and an amphoteric surfactant.

[0142] As the surfactant, from the viewpoint of dispersion stability, it is preferable to use a nonionic surfactant and a cationic surfactant or an amphoteric surfactant in combination, or an anionic surfactant alone, and it is preferable to use a nonionic surfactant and a cationic surfactant in combination.

[0143] A mass ratio of the nonionic surfactant with respect to the cationic surfactant is preferably from 97/3 to 40/60.

[0144] In a specific combination of a nonionic surfactant and a cationic surfactant, since the total amount thereof with respect to the polymer (A) (100% by mass) can be reduced, the adverse effects on the water repellency of the article can be reduced.

[0145] As the nonionic surfactant, one or more types selected from the group consisting of surfactants s^1 to s^6 are preferred.

Surfactant s^1 :

[0146] The surfactant s^1 is a polyoxyalkylene monoalkyl ether, a polyoxyalkylene monoalkenyl ether, or a polyoxyalkylene monoalkapolyenyl ether.

[0147] As the surfactant s^1 , a polyoxyalkylene monoalkyl ether or a polyoxyalkylene monoalkenyl ether is preferred. One type of the surfactant s^1 may be used alone, or two or more types thereof may be used in combination.

[0148] As the alkyl group, the alkenyl group, or the alkapolyenyl group (hereinafter, the alkyl group, the alkenyl group, and the alkapolyenyl group will be collectively referred to as an R^s group), a group having 4 to 26 carbon atoms is preferred. The R^s group may be linear or branched. As a branched R^s group, a secondary alkyl group, a secondary alkenyl group, or a secondary alkapolyenyl group is preferred.

[0149] Specific examples of the R^s group include an octyl group, a dodecyl group, a tetradecyl group, a hexadecyl group, a stearyl group (octadecyl group), a behenyl group (docosyl group), and an oleyl group (9-octadecenyl group).

[0150] As the polyoxyalkylene (hereinafter referred to as POA) chain, a chain in which either one or both of at least two polyoxyethylene (hereinafter referred to as POE) chains or polyoxypropylene (hereinafter referred to as POP) chains are connected is preferred. The POA chain may be a chain composed of one type of POA chain, or may be a chain composed of two or more types of POA chains. When it is composed of two or more types of POA chains, the respective POA chains are preferably linked in a block form.

[0151] As the surfactant s^1 , a compound (s^{11}) is more preferred.

[0152] $R^{10}O[CH_2CH(CH_3)O]_s-(CH_2CH_2O)_rH$ (s^{11}),

[0153] provided that R^{10} is an alkyl group having 8 or more carbon atoms or an alkenyl group having 8 or more carbon atoms, r is an integer of 5 to 50, and s is an integer of 0 to 20.

[0154] When r is 5 or more, it becomes soluble in water and will be uniformly dissolved in an aqueous medium, resulting in favorable permeability of the water repellent composition into an article. When r is 50 or less, hydrophobicity is suppressed and the water repellency is favorable.

[0155] When s is 20 or less, it becomes soluble in water and will be uniformly dissolved in an aqueous medium, resulting in favorable permeability of the water repellent composition into an article.

[0156] When r and s are 2 or more, the POE chain and the POP chain are linked in a block form.

[0157] R^{10} is preferably linear or branched.

[0158] r is preferably an integer of 10 to 30.

[0159] s is preferably an integer of 0 to 10.

[0160] Examples of the compound (s^{11}) include the following compounds. However, the POE chain and the POP chain are linked in a block form.

[0161] $C_{18}H_{37}O[CH_2CH(CH_3)O]_2-(CH_2CH_2O)_{30}H$,

[0162] $C_{18}H_{35}O-(CH_2CH_2O)_{30}H$,

[0163] $C_{16}H_{33}O[CH_2CH(CH_3)O]_5-(CH_2CH_2O)_{20}H$,

[0164] $C_{12}H_{25}O[CH_2CH(CH_3)O]_2-(CH_2CH_2O)_{15}H$,

[0165] $(C_8H_{17})(C_6H_{13})CHO-(CH_2CH_2O)_{15}H$,

[0166] $C_{10}H_{21}O[CH_2CH(CH_3)O]_2-(CH_2CH_2O)_{15}H$.

Surfactant s^2 :

[0167] The surfactant s^2 is a compound having one or more carbon-carbon triple bonds and one or more hydroxyl groups in the molecule.

[0168] As the surfactant s^2 , a compound having one carbon-carbon triple bond and one or two hydroxyl groups in the molecule is preferred.

[0169] The surfactant s^2 may have a POA chain in the molecule. Examples of the POA chain include a POE chain, a POP chain, a chain in which POE chains and POP chains are randomly linked, and a chain in which POE chains and POP chains are linked in a block form.

[0170] As the surfactant s^2 , compounds (s^{21}) to (s^{24}) are preferred.

[0171] $HO-C(R^{11})(R^{12})-C\equiv C-C(R^{13})(R^{14})-OH$ (s^{21})

[0172] $HO-(A^1O)u-C(R^{11})(R^{12})-C\equiv C-C(R^{13})(R^{14})-(OA^2)v-OH$ (s^{22})

[0173] $HO-C(R^{15})(R^{16})-C\equiv C-H$ (s^{23}),

[0174] $HO-(A^3O)w-C(R^{15})(R^{16})-C\equiv C-H$ (s^{24}).

[0175] Each of A^1 to A^3 is an alkylene group.

[0176] Each of u and v is an integer of 0 or more, and $(u+v)$ is an integer of at least 1.

[0177] w is an integer of at least 1.

[0178] When each of u , v and w is 2 or more, each of A^1 , A^2 and A^3 may be the same or different.

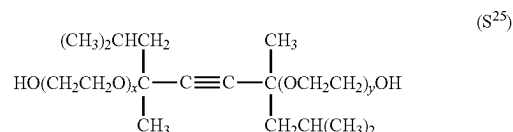
[0179] The POA chain is preferably a POE chain, a POP chain, or a chain containing a POE chain and a POP chain. The number of repeating units in the POA chain is preferably from 1 to 50.

[0180] Each of R^{11} to R^{16} is a hydrogen atom or an alkyl group.

[0181] As the alkyl group, an alkyl group having 1 to 12 carbon atoms is preferred, and an alkyl group having 1 to 4 carbon atoms is more preferred. Examples of the alkyl group include a methyl group, an ethyl group, a propyl group, a butyl group, and an isobutyl group.

[0182] As the compound (s^{22}), a compound (s^{25}) is preferred.

[Chemical Formula 5]



[0183] However, each of x and y is an integer of 0 to 100.

[0184] One type of the compound (s^{25}) may be used alone, or two or more types thereof may be used in combination.

[0185] As the compound (s^{25}), a compound in which x and y are 0, a compound in which an average of the sum of x and y is from 1 to 4, or a compound in which an average of the sum of x and y is from 10 to 30 is preferred.

Surfactant s^3 :

[0186] The surfactant s^3 is a compound in which a POE chain is linked to a POA chain having two or more oxalkylenes of three or more carbon atoms connected in series, and both ends are hydroxyl groups.

[0187] As the above POA chain, either one or both of polyoxytetramethylene (hereinafter referred to as POT) and POP chains are preferred.

[0188] As the surfactant s^3 , a compound (s^{31}) or a compound (s^{32}) is preferred.

[0189] $HO(CH_2CH_2O)_{g1}(C_3H_6O)_t(CH_2CH_2O)_{g2}H$ (s^{31})

[0190] $HO(CH_2CH_2O)_{g1}(CH_2CH_2CH_2CH_2O)_t(CH_2CH_2O)_{g2}H$ (s^{32}).

[0191] $g1$ is an integer of 0 to 200.

[0192] t is an integer of 2 to 100.

[0193] $g2$ is an integer of 0 to 200.

[0194] When $g1$ is 0, $g2$ is an integer of at least 2. When $g2$ is 0, $g1$ is an integer of at least 2.

[0195] $-C_3H_6-$ may be $-CH(CH_3)CH_2-$, $-CH_2CH(CH_3)-$, or a mixture of $-CH(CH_3)CH_2-$ and $-CH_2CH(CH_3)-$.

[0196] The POA chain is in a block form.

[0197] Examples of the surfactant s^3 include the following compounds.

[0198] $HO-(CH_2CH_2O)_{15}-(C_3H_6O)_{35}-(CH_2CH_2O)_{15}H$,

[0199] $HO-(CH_2CH_2O)_8-(C_3H_6O)_{35}-(CH_2CH_2O)_8H$,

[0200] $HO-(CH_2CH_2O)_{45}-(C_3H_6O)_{17}-(CH_2CH_2O)_{45}H$,

[0201] $HO-(CH_2CH_2O)_{34}-(CH_2CH_2CH_2CH_2O)_{28}-(CH_2CH_2O)_{34}H$.

Surfactant s^4 :

[0202] The surfactant s^4 is a compound having an amine oxide moiety in the molecule.

[0203] As the surfactant s^4 , a compound (s^{41}) is preferred.

[0204] $(R^{17})(R^{18})(R^{19})N(\rightarrow O)$ (s^{41}).

[0205] Each of R^{17} to R^{19} is a monovalent hydrocarbon group.

[0206] In the present specification, a surfactant having an amine oxide ($N\rightarrow O$) is treated as a nonionic surfactant.

[0207] One type of the compound (s^{41}) may be used alone, or two or more types thereof may be used in combination.

[0208] As the compound (s^{41}), a compound (s^{42}) is preferred from the viewpoint of dispersion stability of the polymer.

[0209] $(R^{20})(CH_3)_2N(\rightarrow O)$ (s^{42}).

[0210] R^{20} is an alkyl group having 6 to 22 carbon atoms, an alkenyl group having 6 to 22 carbon atoms, a phenyl group to which an alkyl group having 6 to 22 carbon atoms is bonded, or a phenyl group to which an alkenyl group having 6 to 22 carbon atoms is bonded. R^{20} is preferably an alkyl group having 8 to 22 carbon atoms, or an alkenyl group having 8 to 22 carbon atoms.

[0211] Examples of the compound (s^{42}) include the following compounds.

[0212] $[H(CH_2)_{12}](CH_3)_2N(\rightarrow O)$,

[0213] $[H(CH_2)_{14}](CH_3)_2N(\rightarrow O)$,

[0214] $[H(CH_2)_{16}](CH_3)_2N(\rightarrow O)$,

[0215] $[H(CH_2)_{18}](CH_3)_2N(\rightarrow O)$.

Surfactant s^5 :

[0216] The surfactant s^5 is a polyoxyethylene mono(substituted phenyl) ether condensate or a polyoxyethylene mono(substituted phenyl) ether.

[0217] As the substituted phenyl group, a phenyl group substituted with a monovalent hydrocarbon group is preferred, and a phenyl group substituted with an alkyl group, an alkenyl group or a styryl group is more preferred.

[0218] The surfactant s^5 is preferably a polyoxyethylene mono(alkylphenyl)ether condensate, a polyoxyethylene mono(alkenylphenyl)ether condensate, a polyoxyethylene mono(alkylphenyl)ether, a polyoxyethylene mono(alkenylphenyl)ether, or a polyoxyethylene mono[(alkyl)(styryl)phenyl]ether.

[0219] Examples of the polyoxyethylene mono(substituted phenyl) ether condensate or polyoxyethylene mono(substituted phenyl) ether include a formaldehyde condensate of a polyoxyethylene mono(nonylphenyl) ether, a polyoxyethylene mono(nonylphenyl) ether, a polyoxyethylene mono(octylphenyl) ether, a polyoxyethylene mono(oleylphenyl) ether, a polyoxyethylene mono[(nonyl)(styryl)phenyl]ether, and a polyoxyethylene mono[(oleyl)(styryl)phenyl]ether.

Surfactant s^6 :

[0220] The surfactant s^6 is a fatty acid ester of a polyol.

[0221] The polyol represents glycerin, sorbitan, sorbit, a polyglycerin, a polyethylene glycol, a polyoxyethylene glyceryl ether, a polyoxyethylene sorbitan ether or a polyoxyethylene sorbit ether.

[0222] Examples of the surfactant s^6 include a 1:1 (molar ratio) ester of stearic acid and polyethylene glycol, a 1:4 (molar ratio) ester of an ether of sorbit and polyethylene glycol and oleic acid, a 1:1 (molar ratio) ester of an ether of

polyoxyethylene glycol and sorbitan and stearic acid, a 1:1 (molar ratio) ester of an ether of polyethylene glycol and sorbitan and oleic acid, a 1:1 (molar ratio) ester of dodecanoic acid and sorbitan, a 1:1 or 2:1 (molar ratio) ester of oleic acid and decaglycerin, and a 1:1 or 2:1 (molar ratio) ester of stearic acid and decaglycerin.

Surfactant s^7 :

[0223] When the surfactant contains a cationic surfactant, the surfactant s^7 is preferred as the cationic surfactant.

[0224] The surfactant s^7 is a substituted ammonium salt.

[0225] As the surfactant s^7 , an ammonium salt in which one or more hydrogen atoms bonded to a nitrogen atom are substituted with an alkyl group, an alkenyl group, or a POA chain having a hydroxyl group at the end is preferred, and a compound (s^{71}) is more preferred.

[0226] $[(R^{21})_4N^+]\cdot X^-$ (s^{71}).

[0227] R^{21} is a hydrogen atom, an alkyl group having 1 to 22 carbon atoms, an alkenyl group having 2 to 22 carbon atoms, or a POA chain having a hydroxyl group at the end. The four R^{21} groups may be the same or different, but not all of the four R^{21} groups are hydrogen atoms at the same time.

[0228] R^{21} is preferably a long-chain alkyl group having 6 to 22 carbon atoms, or a long-chain alkenyl group having 6 to 22 carbon atoms.

[0229] When R^{21} is an alkyl group other than a long-chain alkyl group, R^{21} is preferably a methyl group or an ethyl group.

[0230] When R^{21} is a POA chain having a hydroxyl group at the end, the POA chain is preferably a POE chain.

[0231] X^- is a counter ion.

[0232] X^- is preferably a chlorine ion, an ethyl sulfate ion or an acetate ion.

[0233] Examples of the compound (s^{71}) include monostearyl trimethyl ammonium chloride, monostearyl dimethyl monoethyl ammonium ethyl sulfate, mono(stearyl) monoethyl di(polyethylene glycol) ammonium chloride, monofluorohexyl trimethyl ammonium chloride, di(tallow alkyl) dimethyl ammonium chloride, and dimethyl monococonut amine acetate.

Surfactant s^8 :

[0234] When the surfactant contains an amphoteric surfactant, the surfactant s^8 is preferred as the amphoteric surfactant.

[0235] The surfactant s^8 is an alanine, an imidazolinium betaine, an amidobetaine, or an acetic acid betaine.

[0236] As a hydrophobic group included in the surfactant s^8 , a long-chain alkyl group having 6 to 22 carbon atoms, or a long-chain alkenyl group having 6 to 22 carbon atoms is preferred.

[0237] Examples of the surfactant s^8 include dodecyl betaine, stearyl betaine, dodecyl carboxymethyl hydroxyethyl imidazolinium betaine, dodecyl dimethylaminoacetic acid betaine and fatty acid amidopropyl dimethylaminoacetic acid betaine.

Surfactant s^9 :

[0238] The surfactant s^9 may be used as a surfactant.

[0239] The surfactant s^9 is a polymeric surfactant composed of a block copolymer or random copolymer of a

hydrophilic monomer and a hydrocarbon-based hydrophobic monomer, or a hydrophobically modified product of a hydrophilic copolymer.

[0240] Examples of the surfactant s^9 include a block or random copolymer of polyethylene glycol (meth)acrylate and a long-chain alkyl acrylate, a block or random copolymer of vinyl acetate and a long-chain alkyl vinyl ether, a block or random copolymer of vinyl acetate and a long-chain alkyl vinyl ester, a polymeric material of styrene and maleic anhydride, a condensate of polyvinyl alcohol and stearic acid, a condensate of polyvinyl alcohol and stearyl mercaptan, a condensate of polyallylamine and stearic acid, a condensate of polyethyleneimine and stearyl alcohol, methylcellulose, hydroxypropyl methylcellulose, and hydroxyethyl methylcellulose.

[0241] Examples of commercially available products of the surfactant s^9 include MP Polymers (article numbers: MP-103, MP-203) manufactured by Kuraray Co., Ltd., SMA resins manufactured by Elf Atochem Inc., Metolose manufactured by Shin-Etsu Chemical Co., Ltd., and Epomin RP manufactured by Nippon Shokubai Co., Ltd.

[0242] As for the combination of surfactants, from the viewpoints of excellent water repellency and durability of the water repellent composition and stability of the obtained emulsion, a combination of the surfactant s^1 , the surfactant s^2 , and the surfactant s^7 , a combination of the surfactant s^1 , the surfactant s^3 , and the surfactant s^7 , or a combination of the surfactant s^1 , the surfactant s^2 , the surfactant s^3 , and the surfactant s^7 is preferred, and the above combination in which the surfactant s^7 is the compound (s^{71}) is more preferred. The total amount of surfactants is preferably from 1 to 10 parts by mass, and more preferably from 1 to 7 parts by mass, with respect to the polymer (100 parts by mass).

(Molecular Weight Modifier)

[0243] As the molecular weight modifier, for example, an aromatic compound, a mercaptoalcohol, a mercaptocarboxylic acid or an alkyl mercaptan is preferred, and a mercaptocarboxylic acid or an alkyl mercaptan is more preferred. Examples of the molecular weight modifier include mercaptoethanol, mercaptopropionic acid, n-octyl mercaptan, n-dodecyl mercaptan, tert-dodecyl mercaptan, stearyl mercaptan, and α -methylstyrene dimer ($\text{CH}_2=\text{C}(\text{Ph})\text{CH}_2\text{C}(\text{CH}_3)_2\text{Ph}$, where Ph is a phenyl group), and n-dodecyl mercaptan, tert-dodecyl mercaptan, or stearyl mercaptan is particularly preferred.

(Polymerization Initiator)

[0244] Examples of the polymerization initiator include a thermal polymerization initiator, a photopolymerization initiator, a radiation polymerization initiator, a radical polymerization initiator, and an ionic polymerization initiator, and a radical polymerization initiator is preferred. As the radical polymerization initiator, for example, an azo-based polymerization initiator, a peroxide-based polymerization initiator, or a redox-based initiator can be used depending on the polymerization temperature. As the radical polymerization initiator, an azo-based compound is preferred, and a salt of an azo-based compound is more preferred. Examples of the initiator include an acetic acid salt of 2,2'-azobis[2-(2-imidazolin-2-yl)propane](VA-061, manufactured by Wako Pure Chemical Industries, Ltd.) and 2,2'-azobis(2-aminodi-

nopropane) (NC-32, manufactured by Nippon Chemicals Co., Ltd.). The polymerization temperature is preferably from 30 to 80° C.

(Other Components)

[0245] The water repellent composition of the present embodiment may contain other components, if necessary.

[0246] Examples of other components include a crosslinking agent, a penetrant, a defoamer, a water absorbent, an antistatic agent, an antistatic polymer, an antirelease agent, a texture adjuster, a film-forming aid, a water-soluble polymer (such as polyacrylamide or polyvinyl alcohol), a thermal curing agent (such as a melamine resin, a urethane resin, a triazine ring-containing compound or an isocyanate-based compound), an epoxy curing agent (such as isophthalic acid dihydrazide, adipic acid dihydrazide, sebacic acid dihydrazide, dodecanedioic acid dihydrazide, 1,6-hexamethylenebis(N,N-dimethylsemincarbazide), 1,1,1', 1'-tetramethyl-4,4'-(methylene-di-para-phenylene) disemicarbazide or spiroglycol), a heat curing catalyst, a crosslinking catalyst, a synthetic resin, a fiber stabilizer, and a fine inorganic particle.

[0247] The water repellent composition of the present embodiment may contain, if necessary, a polymer capable of exhibiting water repellency other than the polymer (A) of the present embodiment, a commercially available water repellent, a water repellent compound having no fluorine atom, or the like. Examples of the water repellent compound having no fluorine atom include a paraffin-based compound, an aliphatic amide-based compound, an alkylethylene urea compound, and a silicon-based compound.

[0248] When the present composition contains a crosslinking agent, adhesion to a base material is likely to be improved.

[0249] As the crosslinking agent, an isocyanate-based crosslinking agent, a methylol-based crosslinking agent, a carbodiimide-based crosslinking agent, and an oxazoline-based crosslinking agent are preferred.

[0250] As the isocyanate-based crosslinking agent, an isocyanate-based crosslinking agent having two or more isocyanate groups is preferred.

[0251] Examples of the isocyanate-based crosslinking agent include an aromatic block type isocyanate-based crosslinking agent, an aliphatic block type isocyanate-based crosslinking agent, an aromatic unblocked type isocyanate-based crosslinking agent, and an aliphatic unblocked type isocyanate-based crosslinking agent. The isocyanate-based crosslinking agent is preferably a water-dispersible type agent emulsified with a surfactant or a self-water-dispersible type agent having a hydrophilic group.

[0252] Examples of the methylol-based crosslinking agent include a condensate or precondensate of urea or melamine with formaldehyde, methylol-dihydroxyethylene-urea and a derivative thereof, methylol-ethylene-urea, methylol-propylene-urea, methylol-triazone, a dicyandiamide-formaldehyde condensate, methylol-carbamate, methylol-(meth)acrylamide, and a polymer thereof.

[0253] A carbodiimide-based crosslinking agent is a polymer having a carbodiimide group in the molecule, and is a crosslinking agent that exhibits excellent reactivity with a carboxy group, amino group, or active hydrogen group in an article or the like.

[0254] An oxazoline-based crosslinking agent is a polymer having an oxazoline group in the molecule, and is a

crosslinking agent that exhibits excellent reactivity with a carboxy group in an article or the like.

[0255] Examples of other crosslinking agents include divinyl sulfone, a polyamide and a cationic derivative thereof, a polyamine and a cationic derivative thereof, an epoxy derivative of diglycidyl glycerol or the like, a halide derivative such as (epoxy-2,3-propyl)trimethylammonium chloride and N-methyl-N-(epoxy-2,3-propyl) morpholinium chloride, a pyridinium salt of chloromethyl ether of ethylene glycol, a polyamine-polyamide-epichlorohydrin resin, a polyvinyl alcohol and a derivative thereof, a polyacrylamide and a derivative thereof, and a glyoxal resin-based antiretardant agent.

[0256] When the present composition contains a methylol-based crosslinking agent or a glyoxal resin-based antiretardant agent, it is preferable to contain a catalyst as an additive. Examples of preferred catalysts include an inorganic amine salt and an organic amine salt. Examples of the inorganic amine salt include ammonium chloride. Examples of the organic amine salt include an amino alcohol hydrochloride and a semicarbazide hydrochloride. Examples of the amino alcohol hydrochloride include monoethanolamine hydrochloride, diethanolamine hydrochloride, triethanolamine hydrochloride, and 2-amino-2-methylpropanol hydrochloride.

(Ratio of Each Component)

[0257] When the present composition contains a medium, the content of the medium can be appropriately selected in accordance with the desired solid content concentration of the present composition.

[0258] The solid content concentration of the present composition immediately after production of the present composition is preferably from 5 to 60% by mass, more preferably from 10 to 50% by mass, and still more preferably from 15 to 45% by mass.

[0259] The solid content concentration of the present composition in a case where the present composition is used for treating an article is preferably from 0.01 to 15% by mass, more preferably from 0.1 to 10% by mass, and still more preferably from 0.5 to 8% by mass.

[0260] It should be noted that the solid content concentration is the total content of the polymer (A) and the surfactant in the present composition.

[0261] The ratio of the polymer (A) with respect to the total mass of the present composition is preferably from 0.01 to 60% by mass, more preferably from 0.1 to 50% by mass, and still more preferably from 0.5 to 45% by mass.

[0262] When the present composition contains a surfactant, the content of the surfactant in the present composition is preferably from 1 to 10 parts by mass, more preferably from 1 to 8 parts by mass, and still more preferably from 2 to 7 parts by mass, with respect to 100 parts by mass of the polymer (A).

[0263] When the present composition contains a molecular weight modifier, the content of the molecular weight modifier in the present composition is preferably from 0.1 to 20 parts by mass, more preferably from 0.1 to 10 parts by mass, still more preferably from 0.5 to 9 parts by mass, and even more preferably from 1 to 8 parts by mass, with respect to 100 parts by mass of the polymer (A).

[0264] When the present composition contains a polymerization initiator, the content of the polymerization initiator in the present composition is preferably from 0.01 to 5 parts

by mass, more preferably from 0.05 to 4 parts by mass, and still more preferably from 0.1 to 3 parts by mass, with respect to 100 parts by mass of the polymer (A).

[0265] The content of fluorine atoms with respect to the total mass of the present composition is preferably 0.1% by mass or less, and more preferably 0% by mass. The content of fluorine atoms with respect to the total mass of the present composition can be measured by combustion ion chromatography or the like.

<<Production Method of Non-Fluorinated Polymer>>

[0266] A method for producing a non-fluorinated polymer according to the present embodiment is a production method in which a mixture containing monomer components including the monomer (a), the monomer (b) and the monomer (c) is polymerized in the presence of a surfactant and a polymerization initiator. Further, a water repellent composition is also produced at the same time.

[0267] The monomer components may further contain either one or both of the monomer (d) and the monomer (e).

[0268] Those produced by known production methods can be used as the monomer (a), the monomer (b), the monomer (c), the monomer (d) and the monomer (e), respectively. If available, commercially available products can be used as monomers.

[0269] The total ratio of the monomer (a), the monomer (b) and the monomer (c) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably 70% by mass or more, more preferably from 75 to 98% by mass, and still more preferably from 80 to 97% by mass. When the total ratio of the monomer (a), the monomer (b) and the monomer (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition and the washing durability thereof are excellent.

[0270] When the monomer component contains either one or both of the monomer (d) and the monomer (e), the total ratio of the monomer (a), the monomer (b) and the monomer (c) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 75 to 98% by mass, and more preferably from 80 to 97% by mass.

[0271] The ratio of the monomer (a) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 50 to 90% by mass, more preferably from 55 to 90% by mass, and still more preferably from 60 to 85% by mass. When the ratio of the monomer (a) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the monomer (a) is equal to or less than the above upper limit value, the monomer (b) and the monomer (c) can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0272] The ratio of the monomer (b) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 5 to 30% by mass, more preferably from 5 to 25% by mass, and still more preferably from 8 to 20% by mass. When the ratio of the monomer (b) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the monomer (b) is equal to or less than the above

upper limit value, the monomer (c) and the monomer (a) can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0273] The ratio of the monomer (c) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 0.1 to 20% by mass, more preferably from 0.5 to 15% by mass, and still more preferably from 1 to 10% by mass. When the ratio of the monomer (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the ratio of the monomer (c) is equal to or less than the above upper limit value, the polymerization yield is favorable, the non-fluorinated polymer (A) can sufficiently contain units based on the monomer (a) and the monomer (b), and the water repellency of the article treated with the present composition is further improved.

[0274] The total ratio of the monomer (a) and the monomer (b) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably 55% by mass or more, more preferably from 60 to 98% by mass, and still more preferably from 68 to 96% by mass. When the total ratio of the monomer (a) and the monomer (b) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the total ratio of the monomer (a) and the monomer (b) is equal to or less than the above upper limit value, the monomer (c) can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0275] The total ratio of the monomer (b) and the monomer (c) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 5.1 to 50% by mass, more preferably from 5.5 to 40% by mass, and still more preferably from 6 to 30% by mass. When the total ratio of the monomer (b) and the monomer (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the total ratio of the monomer (b) and the monomer (c) is equal to or less than the above upper limit value, the monomer (a) can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0276] The total ratio of the monomer (c) and the monomer (a) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 50.1 to 95% by mass, more preferably from 55 to 90% by mass, and still more preferably from 60 to 90% by mass. When the total ratio of the monomer (c) and the monomer (a) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the total ratio of the monomer (c) and the monomer (a) is equal to or less than the above upper limit value, the monomer (b) can be sufficiently contained, and the water repellency of the article treated with the present composition is further improved.

[0277] The mass ratio of monomer (a)/monomer (b) contained in the mixture is preferably from 1.6 to 19, more preferably from 2.2 to 18, and still more preferably from 3 to 10. When the mass ratio of monomer (a)/monomer (b) is

equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of monomer (a)/monomer (b) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0278] The mass ratio of monomer (b)/monomer (c) contained in the mixture is preferably from 0.25 to 300, more preferably from 0.3 to 50, and still more preferably from 0.8 to 20. When the mass ratio of monomer (b)/monomer (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of monomer (b)/monomer (c) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0279] The mass ratio of monomer (c)/monomer (a) contained in the mixture is preferably from 0.001 to 0.4, more preferably from 0.005 to 0.27, and still more preferably from 0.01 to 0.17. When the mass ratio of monomer (c)/monomer (a) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of monomer (c)/monomer (a) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0280] The mass ratio of (monomer (a)+monomer (b))/monomer (c) contained in the mixture is preferably from 2.75 to 950, more preferably from 4 to 800, and still more preferably from 6 to 200. When the mass ratio of (monomer (a)+monomer (b))/monomer (c) is equal to or more than the above lower limit value, the water repellency of the article treated with the present composition is further improved. When the mass ratio of (monomer (a)+monomer (b))/monomer (c) is equal to or less than the above upper limit value, polymerization of the monomer components proceeds easily during the production of the polymer (A).

[0281] When the mixture contains the monomer (d), the ratio of the monomer (d) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 0.1 to 15% by mass, more preferably from 0.2 to 10% by mass, and still more preferably from 0.5 to 6% by mass.

[0282] When the mixture contains the monomer (e), the ratio of the monomer (e) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 0.1 to 15% by mass, more preferably from 0.2 to 10% by mass, and still more preferably from 0.5 to 6% by mass.

[0283] When the mixture contains the monomer (d) or the monomer (e), the total ratio of the monomer (d) and the monomer (e) contained in the mixture with respect to the total mass of the monomers constituting the non-fluorinated polymer is preferably from 0.2 to 30% by mass, more preferably from 0.4 to 20% by mass, and still more preferably from 1 to 12% by mass.

[0284] Examples of a method for polymerizing the monomer components include an emulsion polymerization method, a solution polymerization method, a suspension polymerization method, and a bulk polymerization method. Among these, an emulsion polymerization method is preferred. By polymerizing the monomer components by an

emulsion polymerization method, the molecular weights (Mn, Mw) of the polymer (A) can be increased.

[0285] In the emulsion polymerization method, for example, in an emulsion containing a medium, a monomer component, a surfactant, and a polymerization initiator, the monomer component is polymerized. The emulsion may contain a molecular weight modifier, if necessary.

[0286] Examples of the medium, the surfactant, the polymerization initiator and the molecular weight modifier include the same as those described above.

[0287] The ratio of the above molecular weight modifier with respect to 100 parts by mass of all monomers constituting the non-fluorinated polymer is preferably from 0.1 to 20 parts by mass, more preferably from 0.1 to 10 parts by mass, and still more preferably from 0.1 to 5 parts by mass.

[0288] The emulsion can be prepared by mixing the medium, the monomer component, and, if necessary, a surfactant, dispersing the resulting mixture with a homogenizer, a high-pressure emulsifier, or the like, and then adding a polymerization initiator. It should be noted that when the monomer component is a gas, it can be added to the reaction system after the above dispersion.

[0289] The concentration of the monomer component in the emulsion is preferably from 5 to 60% by mass, and more preferably from 10 to 50% by mass. When the concentration of the monomer component in the emulsion is within the above range, the molecular weight of the polymer (A) can be made sufficiently high.

[0290] The content of the surfactant in the emulsion is preferably from 0.1 to 10 parts by mass with respect to 100 parts by mass of the monomer component. When the content of the surfactant is equal to or more than the above lower limit value, the dispersion stability of the emulsion is excellent. When the content of the surfactant is equal to or less than the above upper limit value, the adverse effects due to the surfactant on the water repellency of an article treated with a composition containing the polymer (A) can be reduced.

[0291] The content of the polymerization initiator in the emulsion is preferably from 0.01 to 5 parts by mass with respect to 100 parts by mass of the monomer component. When the content of the polymerization initiator is equal to or more than the above lower limit value, the reaction rate is likely to be high and the polymerization yield is likely to be improved. When the content of the polymerization initiator is equal to or less than the above upper limit value, a polymer having a molecular weight in the desired range is likely to be obtained.

[0292] A dispersion of the polymer (A) can be obtained by polymerizing the monomer component in the emulsion.

[0293] In the dispersion, the polymer (A) is dispersed as emulsified particles in an aqueous medium.

[0294] In the dispersion, the average particle size of emulsified particles of the polymer (A) is preferably from 10 to 1,000 nm, more preferably from 30 to 600 nm, and still more preferably from 50 to 300 nm. When the average particle size is equal to or less than the above upper limit value, the water repellency of the article treated with the emulsified particles of the polymer (A) and the dispersibility of the emulsified particles of the polymer (A) are further improved. When the average particle size is equal to or more than the above lower limit value, the emulsified particles of the polymer (A) are more stable against mechanical shear forces.

[0295] The average particle size of the emulsified particles of the polymer (A) is calculated by the cumulant analysis method from the autocorrelation function obtained by the dynamic light scattering method for a sample obtained by diluting the dispersion of the polymer (A) with water to a solid content concentration of 1% by mass.

[0296] The dispersion of the polymer (A) obtained by polymerizing the monomer component in the emulsion may be used as it is as the present composition, or may be used as the present composition after diluting with an aqueous medium to adjust the solid content concentration. Other components may be further added to the present composition.

<Treatment Method of Water Repellent Composition>

[0297] A treatment method of the present embodiment is a treatment method using the present composition. Any method may be used as long as the present composition can be adhered to an article to be treated, and for example, when the present composition contains a liquid medium, a method of treating the article with the present dispersion by a known method such as coating, impregnation, immersion, spraying, brushing, padding, size press coating, and roll coating, followed by drying can be mentioned.

[0298] The amount of solid content in the water repellent composition to be adhered to the article to be treated is not particularly limited, but in the case of a fiber fabric, for example, it is preferably from 0.1 to 5 g, per 100 g of the fiber fabric.

[0299] The amount of the polymer (A) in the water repellent composition to be adhered to the article to be treated is not particularly limited, but in the case of a fiber fabric, for example, it is preferably from 0.05 to 5 g, per 100 g of the fiber fabric.

[0300] The drying may be performed at room temperature or by heating, and is preferably performed by heating. When heating is performed, the heating temperature is preferably from 90 to 200° C. Further, when the water repellent composition contains a crosslinking agent, it is preferable to cure by heating to a temperature equal to or higher than the crosslinking temperature of the crosslinking agent, if necessary.

<Article>

[0301] The article of the present embodiment is an article treated with the present composition.

[0302] Examples of the article to be treated with the present composition include fibers, fiber fabrics (such as fiber woven fabrics, fiber knitted fabrics, nonwoven fabrics, and raised fabrics), textile products provided with fiber fabrics (such as clothing including ski wear, rainwear, coats, blousons, windbreakers, down jackets, sportswear, work clothing, uniforms, and protective clothing, backpacks, bags, and tents), glass, paper, wood, leather, artificial leather, stone, concrete, ceramics, metals, metal oxides, ceramic products, resin molded products, and porous resins. The porous resins are used, for example, as filters. Examples of materials for the porous resins include polypropylene, polyethylene terephthalate, and polytetrafluoroethylene.

[0303] The article to be treated is preferably a fiber, a fiber fabric, or a textile product provided with a fiber fabric. Although the type of the fiber is not particularly limited, examples thereof include natural fibers such as cotton, wool,

silk, or cellulose; chemical fibers such as polyester, polyamide, acrylic, aramid, rayon, or lyocell; and fibers obtained by using a plurality of these fibers. When a fiber substrate is a nonwoven fabric, examples of the fiber include polyethylene, polypropylene, polyolefin, polyethylene terephthalate, polytetrafluoroethylene, glass, and rayon.

[0304] Although the thickness of the fiber fabric is not particularly limited, it is from 10 m to 5 cm.

(Mechanism of Action)

[0305] In the present composition described above, since the polymer (A) includes the unit (a), the unit (b), and the unit (c), the article treated with the present composition exhibits excellent water repellency. In addition, since the water repellency is less likely to be reduced by washing, the article treated with the present composition exhibits excellent washing durability. A mechanism by which this effect is achieved is thought to be as follows. Since the monomer (a) and the monomer (b) are compounds having one (meth) acryloyl group, it is thought that the polymerization proceeds one-dimensionally. On the other hand, since the monomer (c) is a compound having two or more (meth) acryloyl groups, it is thought that the polymerization proceeds two-dimensionally, and three-dimensionally in some cases. That is, the polymer is expected to have an elaborate structure due to the inclusion of the structural unit based on the monomer (c). It is believed that this elaborate structure contributes to the improvements of water repellency and washing durability.

EXAMPLES

[0306] Hereinafter, the present invention will be described in detail with reference to Examples, but the present invention is not limited thereto. The term “part(s)” means “part(s) by mass”.

[0307] Cases 1 to 11, 16 to 26, and 29 are Examples of the present invention, and Cases 12 to 15, 27, and 28 are Comparative Examples.

<Ratios of Monomer Units>

[0308] The composition of a polymer (ratio of each monomer unit with respect to all the units constituting the polymer) was calculated based on the amount of monomer component charged.

<Molecular Weight of Polymer>

[0309] The molecular weight of the polymer recovered by the following recovery method was measured.

(Recovery of Polymer)

[0310] 20 g of tetrahydrofuran (THF) was added to 10 g of a polymer emulsion and stirred. This mixture was concentrated to 8 g under reduced pressure. The concentrated liquid was added dropwise into a beaker containing 30 g of methanol while stirring to precipitate the polymer. The supernatant was removed by decantation, and the polymer was obtained by vacuum drying at 35° C. overnight.

<Molecular Weight Measurement>

[0311] The recovered polymer was made into a 1% THE solution and passed through a filter with a pore size of 0.45 m to prepare an analytical sample. The number average

molecular weight (Mn) and mass average molecular weight (Mw) of this sample were measured under the following measurement conditions. The measurement conditions are as follows.

[0312] Apparatus: TOSOH EcoSEC HLC-8320GPC, manufactured by Tosoh Corporation;

[0313] Column: Shodex GPC KF-802 (8.0 mm I.D.×300 mm) and KF-806M (8.0 mm I.D.×300 mm) connected in series;

[0314] Measurement temperature: 40° C.;

[0315] Injection volume: 30 µL;

[0316] Flow rate: 1.0 ml/min;

[0317] Eluent: THF;

[0318] Standard sample: EasiCal PS-2, manufactured by Polymer Laboratories.

<Water Repellent Finishing of Fabric>

[0319] An emulsion obtained in a Polymerization Example described below was processed into a fabric using the following method to obtain a water repellent finished fabric. The emulsion and a blocked isocyanate serving as a crosslinking agent (Meikanate TP-10, manufactured by Meisei Chemical Works, Ltd.) were diluted with distilled water to prepare a water repellent composition. A dyed nylon fabric and a polyester fabric were immersed in this water repellent composition and squeezed evenly. The resultant was dried at 110° C. for 90 seconds using a tenter, and then further subjected to a heat treatment at 170° C. for 90 seconds.

<Water Repellency Evaluation (Initial Stage)>

[0320] The test fabric was evaluated for water repellency in accordance with a spraying test of JIS L1092-2009. The water repellency was expressed in five grades from 1 to 5. The higher the score, the better the water repellency. A grade marked with a symbol of +(or -) indicates that the respective properties are slightly better (or worse) than the standard level of the above grade.

<Washing Durability Evaluation (after Washing)>

[0321] The test fabric was repeatedly washed 20 times in accordance with the C₄M water washing method specified in JIS L1930 Annex F [Specification for washing procedures for reference washing machine Type C (pulsator type)]. After washing, it was dried using a tumble dryer and air dried overnight in a room at a room temperature of 25° C. and a humidity of 60%, and then the water repellency described above was evaluated.

<Overall Evaluation>

[0322] For Cases 16 to 29, 3 points were given for 3, 3.2 points were given for 3+, 3.8 points were given for 4-, and 4 points were given for 4, and the total of the initial evaluation and evaluation after washing for the polyester fabric and the initial evaluation and evaluation after washing for the nylon fabric was used as the overall evaluation.

[0323] The names of the compounds described in the present specification are as follows.

Monomer (a):

[0324] BeA: behenyl acrylate

[0325] StA: stearyl acrylate

Monomer (b):

[0326] VdCl: vinylidene chloride

Monomer (c):

[0327] 16HDA: 1,6-hexanediol diacrylate

[0328] 110DDA: 1,10-decanediol diacrylate

[0329] 16HDMA: 1,6-hexanediol dimethacrylate

[0330] TMP3A: trimethylolpropane triacrylate

[0331] TMP3MA: trimethylolpropane trimethacrylate

[0332] DTMP4A: ditrimethylolpropane tetraacrylate

[0333] DPE6A: dipentaerythritol hexaacrylate

[0334] BPA2M: bisphenol A dimethacrylate

[0335] ICN3A: tris(2-acryloyloxyethyl)isocyanurate

[0336] 1353ATA: 1,3,5-triacryloylhexahydro-1,3,5-triazine

[0337] NNMBAA: N,N'-methylenebisacrylamide

[0338] NNMBMA: N,N'-methylenebismethacrylamide

Monomer (d):

[0339] D-BP: 3,5-dimethylpyrazole adduct of isocyanatoethyl acrylate

Monomer (e):

[0340] DMAA: dimethylacrylamide

Surfactant:

[0341] E420: polyoxyethylene oleyl ether (Emulgen 420, (product name) manufactured by Kao Corporation, approximately 13 moles of ethylene oxide adduct)

[0342] P204: ethylene oxide-propylene oxide polymer (Plonon 204 (product name) manufactured by NOF Corporation, ethylene oxide ratio: 40% by mass)

[0343] TMAC: solution of stearyl trimethylammonium chloride (63%), isopropanol (32%), and water (5%) ((product name), manufactured by Lion Specialty Chemicals Co., Ltd.)

Molecular Weight Modifier:

[0344] StSH: stearyl mercaptan

Polymerization Initiator:

[0345] VA-061A: acetic acid salt of 2,2'-azobis[2-(2-imidazolin-2-yl)propane](VA-061 manufactured by Wako Pure Chemical Industries, Ltd.)

Medium:

[0346] DPG: dipropylene glycol

[0347] Water

Polymerization Example 1

<Production of Polymer Emulsion>

[0348] 72 parts by mass of StA, 4 parts by mass of 16HDA, 4 parts by mass of MOI-BP, 2.5 parts by mass of E420, 2.5 parts by mass of P204, 0.5 parts by mass of TMAC, 4 parts by mass of StSH, 155 parts by mass of water, and 30 parts by mass of DPG were placed in a glass beaker and heated at 60° C. for 30 minutes, and then mixed using a homomixer (BIO MIXER manufactured by Nihonseiki Kaisha, Ltd.) to obtain a mixed liquid.

[0349] The above mixed liquid was treated at 40 MPa, while being kept at 60° C., using a high-pressure emulsifier (Mini-lab, manufactured by APV Rannie), to obtain an emulsion. This emulsion was placed in a stainless steel reactor and cooled to 40° C. or lower. On the other hand, 20 parts by mass of VdCl and 0.5 parts by mass of VA061A were added thereto, and after replacing the gas phase with nitrogen, a polymerization reaction was carried out at 60° C. for 15 hours while stirring to obtain a polymer emulsion. The solid content concentration was 36.8% by mass.

Polymerization Examples 2 to 15

[0350] Polymer emulsions of Polymerization Examples 2 to 15 were obtained in the same manner as in Polymerization Example 1, with the exception that the charging ratios were changed to those shown in Table 1. The solid content concentrations of the obtained emulsions are also shown in Table 1.

Cases 1 to 15

[0351] The emulsions obtained in Polymerization Examples 1 to 15 were processed at a solid content of 1% by mass in accordance with the above section entitled ““water repellent finishing of fabric”, and the obtained processed fabrics were evaluated in accordance with their water repellency. The evaluation results are shown in Table 1. In Cases 1 to 11 in which the monomer (a), the monomer (b), and the monomer (c) were used, better water repellency could be achieved compared to those in Cases 12 to 15 in which the monomer (a) was used, but either one or both of the monomer (b) and the monomer (c) were not used.

Polymerization Examples 16 to 29

[0352] Polymer emulsions of Polymerization Examples 16 to 29 were obtained in the same manner as in Polymerization Example 1, with the exception that the charging ratios were changed to those shown in Table 2. The solid content concentrations of the obtained emulsions are also shown in Table 2.

Cases 16 to 29

[0353] The emulsions obtained in Polymerization Examples 16 to 29 were processed at a solid content of 1% by mass in accordance with the above section entitled ““water repellent finishing of fabric”, and the obtained processed fabrics were evaluated in accordance with their water repellency. The evaluation results are shown in Table 2. In Cases 16 to 26 and 29 in which the monomer (a), the monomer (b), and the monomer (c) were used, better water repellency and washing durability of water repellency could

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TABLE 2-continued

	(b)	VdCl	9.6	9.6	9.6	9.6	9.6	9.5	9.6
	(d)	MOI-BP	2	2	2	2	2	2	2
	(e)	DMAA	2	2	2	2	2	2	2
	(c)	16HDA	2	4	2	4	6	2	4
Surfactant		E420	2.5	2.5	25	2.5	25	2.5	2.5
		P204	0.5	0.5	0.5	0.5	0.5	0.5	0.5
		TMAC	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Molecular weight modifier		StSH	3	3	5	5	5	7	7
Polymerization initiator		VA061A	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Medium		DPG	30	30	30	30	30	30	30
		Water	155	155	155	355	155	155	155
Solid content concentration [%]			37.4	37.5	36.4	37.1	36.4	36.9	37.1
Polymer molecular weight	Mn		23,000	—	25,100	—	—	21,800	25,900
	Mw		42,900	—	47,400	—	—	33,100	53,200
Water repellency performance	Polyester	Initial stage	4-	3+	4-	4-	3+	4-	3+
		After washing	4-	4-	3+	3	4-	3+	3+
	Nylon	Initial stage	4-	4-	3+	4-	3+	3+	4
		After washing	4-	4	4-	4-	3+	3+	4-
Overall evaluation			15.2	14.8	14.0	14.4	13.4	13.4	14.2

			Case						
			23	24	25	26	27	28	29
Monomer	(a)	StA	25	23	25	23	28.8	28.8	29
		BeA	55.4	55.4	55.4	55.4	57.6	57.6	43
	(b)	VdCl	9.6	9.6	9.6	9.6	9.6	9.6	20
	(d)	MOI-BP	2	2	2	2	4	4	4
	(e)	DMAA	2	2	2	2	—	—	—
	(c)	16HDA	6	8	6	8	—	—	4
Surfactant		E420	2.5	2.5	2.5	2.5	2.5	2.5	—
		P204	0.5	0.5	0.5	0.5	0.5	0.5	—
		TMAC	0.5	0.5	0.5	0.5	0.5	0.5	—
Molecular weight modifier		StSH	7	7	10	10	1	2	—
Polymerization initiator		VA061A	0.5	0.5	0.3	0.5	0.5	0.5	—
Medium		DPG	30	30	30	30	30	30	—
		Water	155	155	155	155	155	155	—
Solid content concentration [%]			37.3	37.4	38.1	38.0	36.9	36.1	36.8
Polymer molecular weight	Mn		—	—	—	—	27,500	16,700	—
	Mw		—	—	—	—	58,400	33,200	—
Water repellency performance	Polyester	Initial stage	3+	3+	3+	3+	3+	3	4-
		After washing	4-	3+	4-	4-	3+	3	4-
	Nylon	Initial stage	4-	4-	4-	4	3+	3+	5-
		After washing	4-	3+	3+	3+	3	3	4-
Overall evaluation			14.6	13.4	14.0	14.2	12.6	12.2	16.2

1. A water repellent composition comprising:

a non-fluorinated polymer having a structural unit based on a long-chain alkyl (meth)acrylate monomer;
a structural unit based on a vinyl halide monomer; and
a structural unit based on a monomer having two or more (meth)acryloyl groups.

2. The water repellent composition according to claim 1, wherein said long-chain alkyl (meth)acrylate monomer is represented by the following formula (1):



wherein R^1 represents H, CH_3 or a chlorine atom, and R^2 represents an alkyl group having 12 to 30 carbon atoms.

3. The water repellent composition according to claim 2,

wherein said structural unit based on a long-chain alkyl (meth)acrylate monomer includes a structural unit based on a long-chain alkyl (meth)acrylate monomer in which R^2 in said formula (1) has 12 to 18 carbon atoms, and a structural unit based on a long-chain alkyl (meth)acrylate monomer in which R^2 in said formula (1) has 19 to 30 carbon atoms.

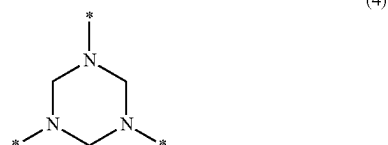
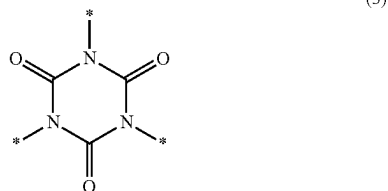
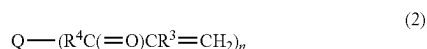
4. The water repellent composition according to claim 3,

wherein a ratio of the structural unit based on a long-chain alkyl (meth)acrylate monomer in which R^2 in said formula (1) has 12 to 18 carbon atoms with respect to

a total amount of said structural unit based on a long-chain alkyl (meth)acrylate monomer is from 1 to 45% by mass.

5. The water repellent composition according to claim 1, wherein said vinyl halide monomer is vinyl chloride or vinylidene chloride.

6. The water repellent composition according to claim 1, wherein said monomer having two or more (meth)acryloyl groups is represented by the following formula (2):



wherein n represents an integer of 2 to 10; R³ represents H, CH₃ or a chlorine atom; R⁴ represents a single bond or a divalent organic group; Q represents a linear or branched alkylene group having 1 to 16 carbon atoms, a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, a trivalent group represented by said formula (3), or a trivalent group represented by said formula (4); a plurality of (R⁴C(=O)CR³=CH₂) moieties may be the same or different, and * represents a bond in said formula (3) and said formula (4), provided that when Q is a linear or branched alkylene group having 1 to 16 carbon

atoms or a residue obtained by removing hydrogen from a hydroxyl group of a polyhydric alcohol, an atom in R⁴ bonded to Q is an atom other than an oxygen atom.

7. The water repellent composition according to claim 1, wherein a ratio of said structural unit based on a long-chain alkyl (meth)acrylate monomer is from 50 to 90% by mass, a ratio of said structural unit based on a vinyl halide monomer is from 5 to 30% by mass, and a ratio of said structural unit based on a monomer having two or more (meth)acryloyl groups is from 0.1 to 20% by mass, with respect to a total amount of structural units of said non-fluorinated polymer.

8. A method for producing a non-fluorinated polymer, the method comprising polymerizing a mixture containing a long-chain alkyl (meth)acrylate monomer, a vinyl halide monomer, and a monomer having two or more (meth)acryloyl groups in the presence of a surfactant and a polymerization initiator,

wherein a ratio of said long-chain alkyl (meth)acrylate monomer is from 50 to 90% by mass, a ratio of said vinyl halide monomer is from 5 to 30% by mass, and a ratio of said monomer having two or more (meth)acryloyl groups is from 0.1 to 20% by mass, with respect to a total mass of monomers constituting said non-fluorinated polymer.

9. The method for producing a non-fluorinated polymer according to claim 8, further comprising polymerizing in the presence of a molecular weight modifier.

10. The method for producing a non-fluorinated polymer according to claim 9,

wherein a ratio of said molecular weight modifier with respect to 100 parts by mass of all monomers constituting said non-fluorinated polymer is from 0.1 to 20 parts by mass.

11. A treatment method using the water repellent composition according to claim 1.

12. An article treated using the water repellent composition according to claim 1.

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